Project Proposal Review

By Milestone 2, we would have already created a working skeleton of the model. This includes training for less loss per loop, testing, and model creation. Ivy will be in charge of creating the testing loop while Anthony will work on the Model class, training, and validation.

Dataset:

When initializing the dataset, the data was already neatly piled into its own type of folders. This made it easier on us because we didn't have to split it ourselves, however, they didn't take into consideration the validation. This meant that we had to split the testing folder into half to make room for validation and testing. When we were running the model to put out the images, we also decided to just use PM2.5 concentration as the model output instead of having it get classified into different classes. We originally were going to use the given code to transform the data into tensors, but we decided to use most of the sample code because it seems like they were needed in order to run the data itself. This includes the transforms, collate functions, and dataloaders.

Model:

We are still keeping the idea of having the images get put into the model and have the model predict a float. However, we decided to ignore classification of the concentration for now, round to the nearest integer, and try to get the model to work at least.

Accuracy Prediction:

Based on rudimentary accuracy for our Model, our current training loss is alright so far due to the fact we have not been able to implement RunPod for training on GPUs yet. However we have been able to connect to a GPU and upload our files, so we will begin implementation! Currently, the percentage loss is noticeable, starting from 24,250% all the way to 7,533%. We

expect better accuracy through larger epoch runs once we are connected to a stronger GPU and are still hoping for double digit loss rate as expected. Over the next week, we will continue to implement more layer modifications to see if we can increase the loss rate more before training on RunPod.

[Output shows Loss *per batch* not per epoch run]

TRAIN OUTPUTS:

Train LOSS: 24250.56640625

Train LOSS: 22451.35546875

Train LOSS: 13049.923828125

Train LOSS: 15118.009765625

Train LOSS: 10119.529296875

Train LOSS: 6564.84912109375

Train LOSS: 8530.1123046875

Train LOSS: 8637.091796875

Train LOSS: 7591.7783203125

Train LOSS: 7169.615234375

Train LOSS: 8710.1796875

Train LOSS: 9220.9443359375

Train LOSS: 8108.390625

Train LOSS: 11755.3330078125

Train LOSS: 7138.65771484375

Train LOSS: 4677.21044921875

Train LOSS: 16245.7080078125

Train LOSS: 12093.4365234375

Train LOSS: 6590.9169921875

Train LOSS: 6560.0869140625

Train LOSS: 7476.13623046875

Train LOSS: 6666.91845703125

Train LOSS: 8204.283203125

Train LOSS: 5050.6279296875

Train LOSS: 5416.48046875

Train LOSS: 6911.498046875

Train LOSS: 7533.5615234375

Weekly Meeting Time/Date:

We are still taking the time to call on Wednesday 8pm-10pm and meeting during office hours on Friday for any in person work. We are thinking about meeting Thursday at 11am before class too, before the demonstration of our model.

Timeline:

For Milestone 0, we both worked on writing the proposal and picking out the dataset together.

For Milestone 1, after picking the dataset, Ivy was able to make visualizations of the images, print out their batches, and create augmentations for further use. Anthony worked mainly on writing the writeup and making the block diagram.

By Milestone 2, we have a working functional model implemented. The training loss is still higher than expected, but we plan to work on that next week before the demonstration. Anthony was in charge of implementing the Model class, and creating the training and validation loss loop. Ivy was in charge of adding the testing loop. Together, we created the concept map to turn in alongside the new proposal and code.

For final submission, we plan on implementing RunPod to store the training weights and biases for easier running in the future. There will be lots of debugging sessions and ways to optimize the model including changing Epoch, layers, learning rate, and batch size. Additionally, we will graph and analyze our models' stats{Loss(y) vs Epoch(x), Accuracy, Precision}. Finally, we will try to practice going through the presentation and making sure the code works.